

MAT 1013: Discrete Mathematics

Introduction

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Course Aims

This course aims to develop a *foundational understanding* of *discrete mathematical structures* essential for mathematics, computer science and related disciplines. Students will explore core concepts in **logic**, **set theory**, **relations**, **functions**, and **number systems**, and *apply formal reasoning* techniques such as **truth tables**, **propositional algebra**, and **quantifiers** to *analyze* and *validate* logical statements.

Emphasis is placed on the *rigorous use of mathematical notation*, *problem-solving strategies*, and the *interpretation of algebraic and graphical representations*.

By the end of the course, students will be equipped to *communicate mathematical arguments clearly* and *apply discrete mathematics principles to model and solve real-world computational problems*.

Learning Outcomes (Overview)

The course is built around six intended learning outcomes focusing on...

- logic and notation,
- logical analysis with truth tables and quantifiers,
- number systems,
- relations and functions,
- graphical/algebraic representations of real functions and sequences,
and
- clear mathematical communication.

Intended Learning Outcomes

At the completion of this course, students will be able to;

- 1 **Explain** fundamental concepts of logic, set theory, relations, and functions, **using** correct mathematical notation.
- 2 **Apply** truth tables, algebra of propositions, and quantifiers to **analyze** and **simplify** logical statements, and to **determine** their validity.
- 3 **Recall basic properties** of number systems (including that of Natural numbers, Integers, Rational numbers, Real numbers and Complex numbers) and **use** these properties (such as Euclidean algorithm, modular arithmetic, prime factorization, absolute value and conjugate of a complex number) to **solve problems** in number systems.
- 4 **Analyze** and **determine** properties of relations and functions, including injectivity, surjectivity, invertibility, and composition.
- 5 **Interpret** and **construct** graphical and algebraic representations of real functions and sequences to model given problems.
- 6 **Communicate** solutions to discrete mathematics problems clearly, **using** structured reasoning and proper mathematical notation.

Recommended Reading

- 1 Rosen, K. H. (2019). Discrete mathematics and its applications (8th ed.). McGraw-Hill
- 2 Epp, S. S. (2019). Discrete mathematics with applications. Cengage Learning.
- 3 Levin, O. (2023). Discrete mathematics: An open introduction (3rd ed)

- This is a new course. Attendance can be very important.
- You are assessed on what we discuss in the class and activities/assignments given to you.

Warning

There can be unannounced quizzes during the lectures, that can contribute to your Continuous Assessment (CA)

Tutorial Session

- Wednesday 5pm to 6pm (Check your timetable)
- Will discuss problems from the lectures and assignments.
- **Weekly Quiz**
- Good place to get your questions answered.

Quizzes and Assignments

- Weekly Quizzes and Assignments
- Tutorial session is accompanied with a weekly Quiz
- There can be unannounced quizzes (popup quizzes) during the lecture.
- Answers to selected problems in the Assignments needs to be submitted.

CONTINUOUS ASSESSMENT

- 1 Quizzes
- 2 Assignments
- 3 Midterm Exam(s)

Contribution to the final grade:
Upto 30%

SUMMATIVE ASSESSMENT

- 1 Final Exam

Contribution to the final grade:
At least 70%

“The only way to learn mathematics is to do
mathematics”

– Paul Halmos